ShopSmart Retail Analytics & AI Implementation

Machine Learning Track — Week 1 | Umar Muhammad | AnalystLab Africa Experience Lab, Batch F

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Part 1 — Machine Learning Fundamentals

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What is Machine Learning?

Machine Learning (ML) is a branch of Artificial Intelligence that enables computers to learn patterns from data and make predictions or decisions, without being explicitly programmed with fixed rules for every scenario. Instead of a developer writing “if X, then Y” for every possible case, an ML model is shown many examples of past data and learns the underlying pattern itself — which it can then apply to new, unseen data.

For ShopSmart, this means instead of manually guessing which orders might be returned, a model can learn from thousands of past orders and predict return likelihood automatically.

Types of Machine Learning

Supervised Learning The model is trained on labeled data — data where the “correct answer” is already known. For example, past ShopSmart orders where we already know the final order total, or whether the order was returned. The model learns the relationship between the input features (e.g. product, quantity, customer) and the known output, so it can predict the output for new orders. This is the type of ML used in this project.

Unsupervised Learning The model works with unlabeled data — there is no known “correct answer.” Instead, it finds hidden structure or groupings on its own. For example, grouping ShopSmart customers into segments based on purchasing behavior, without being told in advance what those segments should be.

Reinforcement Learning The model learns through trial and error, receiving rewards or penalties based on its actions, gradually improving its strategy over time. This is common in robotics and game-playing AI, and less commonly used in typical retail analytics tasks like this one.

Regression vs. Classification

Both are types of supervised learning, distinguished by what kind of answer they predict:

	Regression	Classification
Predicts	A continuous number	A category/class
ShopSmart example	Predicting an order's total value (Order_Total)	Predicting whether an order will be returned (Return_Flag: Yes/No)
Used in this project for	Part 3 (Linear Regression)	Part 4 (Logistic Regression)

The Machine Learning Workflow

1. **Define the business problem** — e.g., "Can we predict which orders are likely to be returned?"
2. **Collect and prepare data** — gather relevant datasets (Orders, Customers, Products, etc.)
3. **Clean the data** — handle missing values, remove duplicates, fix inconsistent formats
4. **Split the data** — divide into a training set (to teach the model) and a test set (to fairly evaluate it on unseen data)
5. **Train the model** — fit a chosen algorithm (e.g. Linear Regression, Logistic Regression) to the training data
6. **Evaluate performance** — measure accuracy using appropriate metrics (MAE, RMSE, R^2 for regression; Accuracy, Precision, Recall, F1 for classification)
7. **Improve/tune** — adjust features or model settings if performance is weak
8. **Deploy/save the model** — store the trained model (e.g. using Joblib) so it can be reused to make predictions on new data without retraining

This workflow is exactly what this Week 1 project follows: preparing ShopSmart's data, building a regression model to predict order value and a classification model to predict returns, evaluating both, and saving them for future use.

Part 5 — Understanding Model Evaluation

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Regression Metrics

MAE (Mean Absolute Error) — the average absolute difference between predicted and actual values, in the same units as the target. Easy to communicate to a business audience: "on average, our order value predictions are off by ₦X." In this project, the regression model (Part 3) achieved an MAE of ₦29,879.88 — about 19.2% of the average order value (₦155,739).

RMSE (Root Mean Squared Error) — similar to MAE, but squares errors before averaging, which penalizes large mistakes more heavily than small ones. Our RMSE (₦40,060.53) was notably higher than our MAE, indicating some predictions were considerably further off than the average case — useful to know if large errors are especially costly to the business.

R^2 Score — the proportion of variance in the target variable explained by the model, ranging from 0 to 1 (and potentially negative for a very poor model). Our model achieved $R^2 = 0.8876$, meaning it explains about 89% of the variation in order totals using legitimate transaction and customer features.

When to use each: MAE is best when you want an intuitive, business-friendly error figure. RMSE is best when large errors are disproportionately costly (e.g. budgeting or inventory decisions where a few very wrong predictions could cause real problems). R^2 is best for comparing how much of the outcome the model explains overall, especially when comparing different models against each other.

Classification Metrics

Accuracy — the percentage of all predictions that were correct. Simple, but highly misleading on imbalanced data. In this project, Return_Flag is imbalanced (only 4.88% of orders were returned). An unbalanced model

achieved 95.12% accuracy — but a Recall of exactly 0%, meaning it never correctly identified a single actual return. It simply learned to always predict “not returned,” which is right 95% of the time by default.

Precision — of all cases the model predicted as positive (e.g. “will be returned”), what percentage actually were positive. Our balanced model had Precision of 4.77% — most flagged orders were false alarms.

Recall — of all actual positive cases, what percentage the model correctly identified. Our balanced model achieved Recall of 46.11%, correctly catching 225 of 488 actual returns in the test set — compared to 0% for the unbalanced model.

F1 Score — the harmonic mean of Precision and Recall, useful as a single balanced summary when both false positives and false negatives matter. Our balanced model’s F1 Score was 0.0864, reflecting the real difficulty of this prediction task with the available features.

When to use each: Accuracy alone is appropriate only when classes are roughly balanced. Precision matters most when false positives are costly (e.g. flagging too many orders for manual review wastes staff time). Recall matters most when missing a true positive is costly (e.g. failing to anticipate a return means no proactive customer service or restocking action). F1 is useful when both error types matter and a single balanced metric is needed for comparison.

Key Finding: Accuracy Can Be Misleading

This project provided a direct, real demonstration of this principle:

Model	Accuracy	Recall
Unbalanced Logistic Regression	95.12%	0.00%
Balanced Logistic Regression (`class_weight='balanced'`)	52.41%	46.11%

The unbalanced model’s high accuracy is an illusion caused by class imbalance — it never identifies a single actual return. The balanced model sacrifices overall accuracy specifically to catch more true positives, which is the more business-relevant trade-off when the goal is actually identifying returns rather than maximizing raw correctness.

Part 7 — Business Questions

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1. Which variables best predict customer spending (Order_Total)? Unit_Price and Quantity are by far the strongest predictors, which makes sense since order value is mathematically derived from these. Beyond the direct transaction details, Customer_Lifetime_Value has the next-largest effect — customers with a higher historical spend tend to place somewhat larger individual orders. Inventory_Status also shows a measurable effect, with orders for well-stocked (high-inventory) items trending slightly lower in value, possibly because popular, high-turnover items tend to be lower-priced.

2. Which variables influence product returns (Return_Flag)? Customer Age Group is the dominant factor — particularly customers aged 56+, who show the strongest positive association with returns. Notably, no product, category, price, or discount feature appears among the top predictors. This indicates that in this dataset, returns are more closely tied to customer demographics than to what was actually purchased or how it was priced.

3. Which products are easiest to predict? The regression model’s errors are driven almost entirely by the Quantity × Unit_Price relationship, which Linear Regression cannot fully capture (it can only combine features additively, not multiplicatively). Orders with typical, mid-range quantities and prices are predicted most accurately; the model struggles most with unusual combinations (e.g. very high quantity with very high unit price), since these produce the largest deviations from the learned linear pattern.

4. Which customer groups generate the highest order values? Differences across Customer_Segment (Silver, Gold, Bronze) and Age Group are small — all segments average within about ₦2,000 of each other (~₦153,000–₦156,000 per order). The 46-55 age group has the highest average order value, but the gap across all groups is not large enough to suggest customer segment or age strongly drives order size on its own.

5. Which stores perform best? ShopSmart Lagos leads in total revenue (₦1.25B), closely followed by Ibadan (₦1.25B) and Abuja (₦1.24B). These top three stores notably outperform Asaba and Kaduna (~₦910M and ₦907M respectively) by roughly 27%, suggesting store location/market size has a more meaningful impact on total performance than customer-level factors do.

6. How could the business use these predictive models? The regression model could support demand and revenue forecasting at the order level once integrated with real-time transaction data. The classification model, while currently weak, highlights a genuine direction for ShopSmart: investigating why older customers (56+) return items more often — this could inform targeted customer service outreach, improved product descriptions/sizing guidance, or return-prevention measures aimed specifically at this age group.

7. What limitations do the models have? The regression model ($R^2 = 0.8876$) cannot fully capture the multiplicative relationship between Quantity and Unit_Price using linear methods alone — a non-linear model (e.g. Random Forest) would likely perform better. The classification model is notably weak (Precision 4.8%, Recall 46.1%, F1 0.086), indicating the available features do not strongly explain why returns happen; this is a limitation of the dataset's feature set, not just model choice — none of the features strongly separate returned from non-returned orders.

8. What additional data would improve predictions? For returns specifically: product condition/quality flags, delivery time or shipping damage reports, customer service interaction history, and specific return reason codes (the Returns dataset has a Return_Reason field not yet incorporated into the model) would likely be far more predictive than demographic data alone. For order value: marketing channel/campaign data and seasonal promotional calendars could help explain variation not captured by transaction details alone.

ML01: Building Your First Machine Learning Models

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Business Problem

ShopSmart Retail Ltd. wants to begin integrating predictive analytics into its retail operations. This project builds two baseline machine learning models using historical retail data (9 linked datasets covering customers, orders, products, stores, returns, inventory, and suppliers):

1. A **regression model** to predict an order's total value (`Order_Total`)
2. A **classification model** to predict whether an order will be returned (`Return_Flag`)

Data Preparation

Nine source datasets (50,000 orders, 5,000 customers, 500 products, 25 stores, 2,500 returns) were merged into a single master dataset using `Order_ID`, `Customer_ID`, `Product_ID`, and `Store_ID` as join keys. All tables were verified to contain zero missing values and zero duplicate rows.

Seven engineered features were added: Order_Month, Order_Quarter, Profit_Margin, Discount_Percentage, Customer_Age_Group, Customer_Lifetime_Value, Days_Since_Last_Purchase, and Inventory_Status. One data quality issue was identified and corrected during preparation: an initial merge with the Inventory table incorrectly joined on both Product_ID and Store_ID, causing 96% of rows to fail to match. Investigation showed inventory is tracked per product company-wide (one row per product), not per store — the join was corrected to use Product_ID only, resolving the issue completely.

Categorical variables (13 columns) were one-hot encoded, producing 71 model-ready features. Numerical features were standardized using `StandardScaler`, fit only on the training data to avoid information leaking from the test set. Data was split 80/20 into training and test sets, stratified on Return_Flag to preserve the 4.88% return rate in both sets given significant class imbalance.

Model Development

Regression (Order_Total): A Linear Regression model was trained on legitimate transaction and customer

features. An initial experiment adding a Quantity × Unit_Price interaction term produced a perfect R^2 of 1.0 — this was recognized as the model trivially reconstructing the known arithmetic of order value, not genuine predictive learning, and was excluded from the final model.

Classification (Return_Flag): A Logistic Regression model was trained with `class_weight='balanced'` to address severe class imbalance (only 4.88% of orders were returned). An unbalanced baseline was also trained for comparison.

Model Evaluation

Model	Metric	Result
Regression	MAE	₦29,879.88
Regression	RMSE	₦40,060.53
Regression	R^2	0.8876
Classification (balanced)	Accuracy	52.41%
Classification (balanced)	Precision	4.77%
Classification (balanced)	Recall	46.11%
Classification (balanced)	F1 Score	0.0864
Classification (unbalanced)	Accuracy	95.12%
Classification (unbalanced)	Recall	0.00%

The unbalanced classification model illustrates a key lesson in ML evaluation: high accuracy (95.12%) can be entirely misleading on imbalanced data — this model never correctly identified a single actual return. The balanced model trades overall accuracy for meaningfully higher recall, which is the more business-relevant outcome when the goal is identifying returns.

Business Insights

- Unit_Price and Quantity are, unsurprisingly, the strongest drivers of order value; Customer_Lifetime_Value and Inventory_Status show smaller secondary effects.
- Return likelihood is most strongly associated with customer age group (particularly 56+), not with product category, price, or discount — suggesting return drivers relate more to customer experience than product/pricing factors.
- Store revenue varies meaningfully by location (Lagos, Ibadan, and Abuja outperform Asaba and Kaduna by ~27%), while customer segment and age group show minimal effect on individual order value.

Recommendations

1. Investigate why customers aged 56+ show a higher return tendency — this may point to a genuine customer experience issue (e.g. product fit, sizing clarity, ease of the return/support process) worth addressing directly.
2. Incorporate the Returns dataset's Return_Reason field and any available product quality/shipping data in future modeling — current features do not explain returns well (Precision 4.8%, F1 0.086), and richer, more targeted data is likely needed.
3. For order value forecasting, consider a non-linear model (e.g. Random Forest or Gradient Boosting) in future iterations, which can better capture the multiplicative relationship between quantity and price than Linear Regression.

Limitations

- The regression model occasionally produces negative predicted order values (a known limitation of unconstrained Linear Regression), which are not meaningful in a real business context.

- The classification model’s low precision (4.8%) means most flagged “at-risk” orders would be false alarms if used operationally as-is — it is not yet reliable enough for automated decision-making, only for directional insight.
- Both models used only the features available in this dataset; several plausible drivers of returns (product quality, delivery experience) were not available and are flagged as recommended additions above.